

4.4.15

AI25BTECH11019 - MENAVATH SAI SANJANA

Question:

Find the sum of the intercepts cut off by the plane $-\mathbf{r} \cdot (2\hat{i} + \hat{j} - \hat{k}) - 5 = 0$ on the three coordinate axes.

Solution:

The general plane can be written as $ax + by + cz + d = 0$, where $\mathbf{r} = x\mathbf{e}_1 + y\mathbf{e}_2 + z\mathbf{e}_3$, and $\mathbf{n} = a\mathbf{e}_1 + b\mathbf{e}_2 + c\mathbf{e}_3$ is the normal.

Intercepts:

$$x\text{-intercept: } -\frac{d}{a}\mathbf{e}_1, \quad (0.1)$$

$$y\text{-intercept: } -\frac{d}{b}\mathbf{e}_2, \quad (0.2)$$

$$z\text{-intercept: } -\frac{d}{c}\mathbf{e}_3. \quad (0.3)$$

Vector sum of intercepts: $-d\left(\frac{1}{a}\mathbf{e}_1 + \frac{1}{b}\mathbf{e}_2 + \frac{1}{c}\mathbf{e}_3\right)$.

Scalar sum of intercepts: $x + y + z = -d\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)$.

For the given plane $2x + y - z + 5 = 0$, we have $a = 2$, $b = 1$, $c = -1$, $d = 5$.

Therefore, $x + y + z = -5\left(\frac{1}{2} + 1 + \frac{1}{-1}\right) = -\frac{5}{2}$.

The sum of the intercepts cut off by the plane is $-\frac{5}{2}$.

Conclusion: Using the basis vectors $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ we obtain the general formula $x + y + z = -d\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)$, valid when the plane $ax + by + cz + d = 0$ meets all three axes. For the given plane the sum of intercepts is $-\frac{5}{2}$. Hence, the sum of intercepts is

$$-\frac{5}{2} - 5 + 5 = -\frac{5}{2}. \quad (0.4)$$

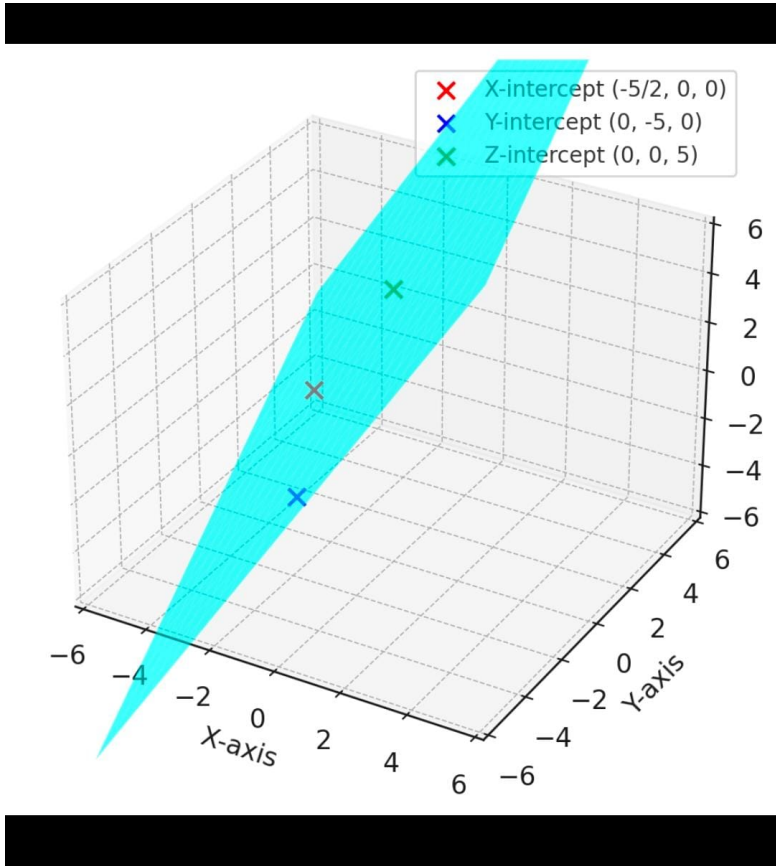


Fig. 0.1